

Multiple Choice

1. **Answer:** C

Options: A) 1 J; B) 500 J; C) -250 J; D) -500 J; E) -100 J

Note: Correct option: C - -250 J

2. **Answer:** A

Options: A) Low temperature and low pressure; B) Low temperature and high pressure; C) High volume and high pressure; D) High temperature and high pressure; E) High volume and low pressure

Note: Correct option: A - Low temperature and low pressure

3. **Answer:** A

Options: A) 3.0000; B) -0.1703; C) -2.1703; D) -0.8297; E) -1.1703

Note: Correct option: A - 3.0000

4. **Answer:** C

Options: A) HCl / NaOH; B) H⁺ / Cl⁻; C) H₃O⁺ / H₂O; D) O₂ / H₂O

Note: Correct option: C - H₃O⁺ / H₂O

5. **Answer:** A

Options: A) 48 ounces of silver; B) 42 ounces of silver; C) 54 ounces of silver; D) 72 ounces of silver; E) 60 ounces of silver

Note: Correct option: A - 48 ounces of silver

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Decrease in the production of n-butane due to formation of ethane and hexane'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1,4-dichlorobenzene'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $T_B = [(2 a) / (Rb)]$ '.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '91.17 nm'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 0.25%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $\text{Br}^-(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NO}_3\text{Br}(\text{aq})$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key